University of Bahrain

College of Information Technology

ITCS285 Database Management Systems

Assignment #4

Normalization

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Given Relation

R3(A, B, C, D, E, F, G, H, I, J, K, L)

Primary Key: **AKL**

Functional Dependencies (FDs):

1. A → B
2. A → D
3. A,E → G
4. B → C
5. D → C
6. E → F
7. E → H
8. H → I
9. J,K,L → E

1. Normalize to 2NF

Reason: Remove partial dependencies (attributes depending on part of the composite key).

Resulting Relations:

Relation	Relation	Key
R1	A, B, D, C	A
R2	A, K, L, E, F, G, H, I, J	A,K,L

2. Normalize to 3NF

Reason: Remove transitive dependencies (non-prime attributes depending on non-key attributes).

Resulting Relations:

Relation	Attributes	Key
R1a	A, B, D	A
R1b	B, C	B
R1c	D, C	D
R2a	E, F	E
R2b	E, H	E
R2c	H, I	H
R2d	A, E, G	A,E
R2e	J, K, L, E	J,K,L
R2f	A, K, L	A,K,L (to preserve original primary key)

3. Normalize to BCNF

Reason: Every determinant is already a superkey, so **BCNF = 3NF**.

Relations remain the same as 3NF:

Relation	Attributes	Key
R1a	A, B, D	A
R1b	B, C	B
R1c	D, C	D
R2a	E, F	E
R2b	E, H	E
R2c	H, I	H
R2d	A, E, G	A,E
R2e	J, K, L, E	J,K,L
R2f	A, K, L	A,K,L